Phase-7 Structural Quant Engine

Remediation Plan

Step 5 of the audit sequence — “prioritise findings by severity and effort.”

Produced by GLM 5.3 (z-ai/glm-5.3) on August 29, 2026. Verified by Claude the same day.

What this document is.

The unedited output of Step 5, preceded by the record of which of its claims were checked against real engine source and what those checks found. The plan is not summarised or reordered.

It is a **plan**, not a finding. It changes no rule and grades no item. The adjudications it opens with are Viktor's, per Roles & Authority, and two of them change the ordering — which is why they sit at position zero rather than being settled along the way.

There are five, not four. Step 5's own Step 0 names four — Item 6's severity, Item 2's Compliant strength, Items 4/12, and the position-sizing question — and raises a fifth at item 9, the choice between halting and degrading on indicator failure. An earlier summary of this document by Claude listed four and dropped the position-sizing one, which propagated into Roadmap Revision 2. Corrected here and in Roadmap Revision 3. *The verbatim plan below was always right; the summary of it was not.*

The reviewer, and what it had.

GLM 5.3 on OpenRouter — the pinned slug, deliberately not the `glm-latest` floating alias, which silently repoints when a successor ships and would make this document's record of which model produced what quietly false. Four attachments: the complete engine source, the frozen audit copy of the Constitution, all four audit runs, and the Engineering Notes.

Independence status: GLM was clean on both the Constitution and source lists before this run. It no longer is. Luna Pro was considered and rejected for this step because it had already seen the Constitution during the hostile review. Still clean on both lists for the Step 8 re-audit: Meta, Mistral, Qwen, Cohere, Amazon, MiniMax.

Verification record

Every claim in the plan that could be checked against the real source was checked. Nine of nine held. This section exists because the Constitution's Evidence discipline requires a finding to be checkable without re-running the reasoning that produced it — and because this project has twice recorded Claude asserting something false about the source.

The plan's claim	Checked against source	Result
`requirements.txt` omits `requests` and `colorama`; declares `ccxt`, which nothing imports	Manifest lists pandas, numpy, matplotlib, ccxt, pandas_ta. `requests` imported at line 1295, `colorama` at 1004. `ccxt` appears exactly once in the whole codebase — in the manifest itself.	**Holds**
The validator must sit before `close_time` is discarded	Line 1369 defines the column in the raw response; line 1373 drops it. Staleness and last-candle-completeness checks need it, and nothing downstream ever sees it.	**Holds** — and is the sharpest single observation in the plan
`engine_version` exists in config and is written nowhere	Defined at config line 260. Total occurrences in the source: one.	**Holds**
The indicator cache can never hit across runs	Key at line 521 is symbol, timeframe, row count and last close, in a per-instance dict. Empty on every fresh process; a new bar produces a new key within one.	**Holds** — deletion is therefore free
mypy does not exist in the repository	Zero occurrences.	**Holds**
Config declares chart height 10 and dpi 150; plotting hardcodes 8 and 200	`CHART_HEIGHT = 10`, `CHART_DPI = 150` at config 356–357; `figsize=(14, 8)` at 4561 and `dpi=200` at 4743.	**Holds**
VWMA must stay — `entry_model` consumes it	VWMA distance scoring at line 3094. Correctly excluded from the deletion list.	**Holds**
The `trend_failure` gate is dead	`recent_struct == "LH"` is exact equality against a column whose values are full phrases such as "BEARISH SWING SEQUENCE (LH-LL)". Never matches. Attributed by the plan to Run 1, which found it and described the mechanism correctly.	**Holds**
"23 commits and zero tags (checked)"	Traced to the Engineering Notes, which record exactly that. A legitimate citation of supplied material, not a fabricated verification.	**Holds**

Two things the verification found that the plan did not

The dead-gate defect is a class, not an instance. The codebase contains two comments acknowledging previous occurrences of the same shape — a gate comparing against a string literal no producer ever emits. One notes a check for `STRUCTURE_REGIME`, "which never existed"; the other notes that "BULLISH STRUCTURE" and "BEARISH STRUCTURE" "were never produced by anything." Both were found and commented rather than turned into a check. That makes `trend_failure` the third instance, and argues for a class-level guard in the harness — assert that every string literal compared against a column belongs to that column's producible set — rather than a third individual repair.

A near-miss on Claude's side, recorded rather than tidied away. Claude was one sentence from reporting that the plan understated the `trend_failure` fix, on the grounds that the `STRUCTURE` column is never assigned anywhere. It is — via a `.loc` form that the first search pattern did not match. That would have been the third false assertion of absence on this project, after the Layer 5 claim and the withdrawn accusation against Run A's auditor. An exhaustive re-search caught it before it reached Viktor. The plan's mechanism was right; the correction would have been wrong.

One real discrepancy.

The plan says “the six logged runtime failure classes.” The runtime log holds seven distinct classes across nine occurrences. An undercount by one, immaterial to the sequence, recorded because unrecorded small errors are how large ones become invisible.

What verification cannot establish.

Every check above confirms that a claim about the source is true. None of them confirms that the *ordering* is right — that is a judgment, and it is reviewed by using the plan, not by grepping. The plan makes this point about itself: its own tests would prove a change matched its intention, never that the intention was correct. Only the independent re-audit at item 16 closes that gap.

The plan, verbatim

Unedited from here to the end of the document, including its three self-flagged correlation warnings — the places where the reviewer notes that its own reasoning may share habits with the model family that built the engine.

The Sequence

The ordering logic, in one breath: **make the audited state recoverable and the engine startable → build the measurement apparatus the findings' own Verification fields presuppose → strip dead complexity → fix the three confirmed Criticals → reshape contracts, risk semantics, and output truthfulness → the backtest on-ramp → independent re-audit.** Severity decides membership in the gate sets, not position; position is set by verifiability prerequisites and by which edits share a seam.

Step 0 — Adjudications and registrations *(process, no code)*

- **Position:** First, because two open disputes have ordering consequences (Item 6's severity can swap positions 12 and 13; Items 4/12 dissolve if repaired) and because the two unrated harness findings must enter the formal record now or be lost the way soft-rated defects already were. Ask Viktor to rule on: Item 6's severity, Item 2's Compliant strength, Items 4/12, and the position-sizing display question Item 14's finding itself defers to a constitutional decision.
- **Depends on:** nothing. **Unblocks:** stable ordering of 12/13; the minimal gate set (Q5).
- **Effort:** small.
- **Risk of doing:** none material; rulings are reversible, dated entries.
- **Risk of not:** positions 12/13 stay conditional; unrated defects ride informally and get dropped.

Phase A — Floor

1. T3-7 — Known-good checkpoints

- **Position:** Before any code changes: Tier 3 requires a preserved state before significant modification, and re-audit diffs must compare against the exact code the findings describe. The repo has 23 commits and zero tags (checked) — this is one tag on the audited commit (commit anything dirty first).
- **Depends on:** nothing. **Unblocks:** every later item's rollback floor.
- **Effort:** small.
- **Risk of doing:** none — a tag is metadata.
- **Risk of not:** no return point; the first bad change becomes archaeology.

2. T2-6 — Controlled dependencies *(severity-corrected: fresh clone does not start)*

- **Position:** Second because nothing downstream is verifiable from a clean environment until the engine runs. Two mechanisms in the source: `requirements.txt` omits `requests` (which `data_fetcher` imports, so `main.py`'s first import fails), and `main.py` constructs a `FileHandler` on `Logs/` at import time, before any `makedirs` — a clone without `Logs/` dies before `main()` executes. Fix both: correct the manifest (add `requests`; declare or document `colorama`; drop `ccxt` — declared, never imported, execution-capable), and create directories before configuring logging.
- **Depends on:** 1. **Unblocks:** fresh-clone verification of everything after; the harness's clone check becomes the regression test.

- **Effort:** small.
- **Risk of doing:** manifest changes can shift installed versions and thus numeric output — detected by the smoke/golden tests once they exist, and by recording the environment lock alongside the pinned dataset.
- **Risk of not:** the published artifact stays unstartable; every fix is verified only on the author's machine — exactly the "soft" rating the harness corrected.
- Note: the rubric caps Tier 2 at Moderate; record the correction as an annotation (operational impact: total blocker). Same severity-versus-impact divergence as Item 6's dispute.

Phase B — Apparatus

3. Pinned-data path and archived datasets *(prerequisite; closes T3-5's Unknown as a side effect)*

- **Position:** The apparatus starts here because the repository cannot produce two comparable runs: `load_csv` exists but nothing wires it in — `get_tf` always fetches live MEXC, so even T2-4's prescribed verification ("change a knob, rerun, compare") is impossible today. Add a source-injection point on `get_tf` covering all three series the engine fetches (base, macro 1d, BTC context — otherwise runs stay network-dependent and nondeterministic); pin clean, closed-candle pulls; record hashes and origin. The Constitution says raw-pull retention is a decision the audit should force — make it here.
- **Depends on:** 2. **Unblocks:** items 4 and 7; the verification clauses of Items 3, 11, 13, T2-4.
- **Effort:** small-medium.
- **Risk of doing:** a `get_tf` signature change touches call sites — contained, caught by the suite.
- **Risk of not:** every later fix is verified by eyeball on data that changes per run.

4. T3-3 / T3-4, stage 1 — commit and extend the harness

- **Position:** The harness exists outside the 19-file source and outperformed four graded passes on runtime facts — bring it in as the suite skeleton: import/compile checks (every one of the six logged runtime failure classes would have been caught), a smoke run on pinned data, and grep guards for `ccxt`/credential patterns (which also serve Item 18's re-checks). No CI exists in the repo — do not assume it; a locally green suite is the standard. Begin commit-per-change with dated notes here: the remediation itself then generates the evidence that can later close T3-1/T3-2/T3-8.
- **Depends on:** 3. **Unblocks:** invariance checks for 5; fixture and fault-injection tests for 8/9; the golden pin at 7.
- **Effort:** medium.
- **Risk of doing:** a suite can enshrine current-wrong behavior — mitigated by pinning goldens only *after* cleanup (5/6) and treating mismatches as reviewable diffs.
- **Risk of not:** Step 7 of the ratified sequence (regression-test each fix) is impossible; the project's own history — a rename that broke fourteen modules, six runtime-caught syntax/import errors — repeats.

5. Item 16 — delete unconsumed complexity

- **Position:** Before the golden pin (the baseline shouldn't enshrine dead code) and before Items 13/8 (every deleted fallback is one fewer fabrication path needing honest semantics; deletion is the compliant remedy — Q4). Delete: the Bollinger trio, KAMA and its slope, DIP/DIM, Typical_Price, the unreachable VWAP plot branch, `compute_exit` (re-source `current_price` directly; exit_watch already owns the advisory role), the dead TARGET-HIT color branches in the panel. Riders from the verified blind run: the unreachable hysteresis branches in `_detect_regime` (Item 12's own remark), and the stale roadmap comments that describe a code state that no longer exists.
- **Depends on:** 3, 4. **Unblocks:** smaller Item 13/8 scope; T2-4 without wiring dead knobs; a clean baseline.
- **Effort:** small-medium.
- **Risk of doing:** deleting something with an undocumented consumer (`live_trading` reads `exit.action`) — detected by grep, the import/compile suite, and a decision-object invariance diff on pinned data. That diff *is* the verification: deleting unconsumed code is provable by output-invariance.

- **Risk of not:** every later change carries dead surface; Item 16 stays Non-compliant.
- *Correlation flag:* deletion is my instinct and also the register's Item 16/T4-6 instinct, written by the same model family. The guards are the invariance proof, git reversibility, and re-audit.

6. T2-1 + cache removal — shared-frame aliasing *(dissolves the Items 4/12 dispute)*

- **Position:** Same disease as T2-1: modules mutating frames they don't own. `calculate_dynamic_bias` rewrites caller columns in place; the indicator cache mutates cached frames on its only reachable hit path while never hitting across runs at all (fresh process; the key embeds the last close). Deleting the caches is cheaper than repairing the key (Q4) and removes the stale-serving hazard Claude's Item 4/12 dispute names — the dispute dissolves by repair rather than adjudication. Also protects the golden tests from in-process contamination.
- **Depends on:** 4. **Unblocks:** a trustworthy golden baseline; the 4/12 ruling becomes trivial.
- **Effort:** small.
- **Risk of doing:** none material — recomputation at 450 bars is trivial and the cache is dead weight in every real usage pattern.
- **Risk of not:** cached-frame mutation can corrupt later analyses on the `live_trading` singleton path; the dispute stays open.

7. T3-4, stage 2 — golden baseline

- **Position:** Captured after cleanup (5/6) and before any semantic change (8+), so every later delta is attributable to exactly one controlled change. Pin decision-object fields on the pinned dataset; seed/delete the C3 state file per test; exclude chart PNGs from goldens (unstable hashes); force any uninjected fetch to fail deterministically (note: the `data_fetcher` singleton binds `base_url` at import — patch the instance, not just config).
- **Depends on:** 3–6. **Unblocks:** verification of 8–13.
- **Effort:** small.
- **Risk of doing:** goldens freeze behavior someone later wants changed — that is what reviewed diffs are for.
- **Risk of not:** Items 11/13 fixes are unprovable as "changed exactly as intended."

Phase C — The Criticals

8. Item 3 — Data Integrity (Critical)

- **Position:** First Critical because it is upstream: no downstream number is trustworthy while garbage can enter silently, and its clean-path invariance ("the validator changes nothing on clean data") is assertable against the golden baseline — a strong, cheap regression property. Scope: a validation gate in `fetch_ohlc` *before* the `close_time` column is discarded (gaps, duplicates, ordering, impossible prices, staleness, volume sanity, last-candle completeness — the raw response carries `close_time` today and throws it away); reject-not-fabricate at ingestion; macro-timeframe failure renders "UNAVAILABLE," never "NEUTRAL" — a fabricated neutral is a directional vote at 10% weight. Rider from the blind run: whitelist valid intervals so the silent `1h` failure dies. Decide and record the forming-candle policy; pin datasets with a closed last candle so the invariance check stays clean.
- **Depends on:** 3, 4, 7. **Unblocks:** a release-gate item; honest inputs for 9/11; a backtest-gate item.
- **Effort:** medium.
- **Risk of doing:** an over-strict validator rejects marginal-but-usable data and availability drops — the fail-safe direction, but detectable by running the full symbol/timeframe scan and reviewing rejection reasons, not by assumption. The corrupted-data fixtures are themselves the new tests (hypothesis stated before the fix, per Tier 3).
- **Risk of not:** garbage becomes confident panels — the release gate's stated reason to exist.

9. Items 13 + 8 — Fail Safely / Epistemic Honesty *(merged defect: fabricated fallbacks)*

- **Position:** Second Critical; depends on 8's fixtures and is shrunk by 5's deletions. Delete the fabrication constants (RSI 50, ADX 25, ATR = 2% of price, SuperTrend = close, ST_Direction = 1.0, the trend_health 50.0 diets, `risk_model`'s direction-blind except-return) and let failures propagate into the existing error-decision path — the "halt output" branch of Item 13's invariant. Log every would-be fallback. *Trade-off flagged:* the auditor's Required action names the flag-and-degrade route (keep computing, mark degraded, cut confidence); halting is cheaper, adds no machinery, and satisfies the invariant's letter — but my preference for the simple branch is a habit I share with the model family that built this engine, so the choice goes to Viktor at step 0; the degraded variant remains available as a later amendment. Rider (verified in the blind run): the `trend_failure` gate is dead — STRUCTURE never contains "LH"/"LL". Either wire it to the real swing sequence (a repair: the data exists and blocking entries on structural failure is the engine's stated job; decisions will change on downtrending fixtures, which is the point, documented as a golden delta) or delete it (simpler; removes a false impression of protection). I recommend wiring; same correlation flag, same visibility to Viktor.

- **Depends on:** 4, 5, 7, 8. **Unblocks:** a release-gate item; honest confidence for 11.
- **Effort:** medium.
- **Risk of doing:** the engine halts where it used to fabricate — availability drops by design; detect via halt-rate across the scan. Wiring `trend_failure` materially changes entry gating — intended, but it must be reviewed as a golden delta, not merged silently.
- **Risk of not:** a broken SuperTrend keeps injecting a permanent +15-point bullish vote into bias — the most dangerous single live-path defect in the audit.

10. T2-3 — decision-object contract

- **Position:** Between the shape-neutral Criticals (8/9 use the existing error path; no new fields) and the reshaping clusters (11–13 add, remove, and rename fields). Declaring the schema now means every subsequent field change is checked against consumers — the guard whose absence once broke fourteen modules. mypy does not exist in the repository (checked); adding it is apparatus belonging to this item. *Defensible alternative:* do Item 11 first and open the release gate one mechanical item sooner; rejected because renaming without a net is this codebase's demonstrated failure mode.
- **Depends on:** 7. **Unblocks:** safe field work in 11–13.
- **Effort:** medium.
- **Risk of doing:** premature rigidity — the schema fights later fixes; mitigate by versioning it with each controlled change rather than freezing it.
- **Risk of not:** 11–13's field edits land unguarded; consumers break silently on rename — T2-3's own evidence says this already happened once.

11. Item 11 — No Circular Reasoning *(+ Item 10(a) rendering; merged root defect)*

- **Position:** Third Critical, after 8/9 so the score being de-circularized is computed on validated, non-fabricated data (otherwise the golden deltas conflate two causes). Remove the `trend_health` term from `_compute_confidence` (it is already inside `bias_strength` at 0.30 weight); rebase `validation_score` off risk-relevant inputs or fold it to one factor — this edit also discharges Item 14(b)'s gate-rebasing requirement (one controlled change, two findings, shared seam); delete the redundant MOMENTUM-number and Current-Market renderings rather than inventing distinct underlying metrics (inventing is new-feature work; T4-5 sides with removal). Coupling rule: the reason strings change in the same commit — prose describing the old formula is an Item 8 regression the moment the number changes.
- **Depends on:** 7–10. **Unblocks:** a release-gate item; Item 14's remainder consumes the rebased gate.
- **Effort:** medium.
- **Risk of doing:** the honest confidence ceiling drops and decisions go WAIT-heavy — an intended shift to be reviewed via scan-run distribution, not rubber-stamped. A wrong ablation (removing the wrong term) is caught by the ablation test: perturb trend_health, assert confidence moves exactly once.
- **Risk of not:** one signal keeps presenting itself as four independent confirmations — the audit's strongest finding, and the one that most inflates whatever a future backtest measures.

Phase D — Shape, truth, configuration

12. Items 5 + 6 — Reproducibility / Traceability *(+ Item 7 label; harness text riders)*

- **Position:** Sequenced here under the stricter reading of the Item 6 severity dispute (the project's own DEFECT-row precedent: unresolved → stricter governs); if Viktor adjudicates Major, this swaps with 13 — both are small. Implement the decision log the panel already claims: append the full decision object plus fingerprint (last-candle timestamp, row count, source endpoint, `engine_version` — which exists in config and is currently written nowhere — config snapshot); make the "Trade logged" line name a file that provably exists; fix the chart-path line (it renders "AI Risk chart saved to None" because the key exists with value None, defeating the `.get` default); add Item 7's "computationally validated, empirically unvalidated" label adjacent to BTC-ADJUSTED CONFIDENCE; replace the hardcoded AERO with the run's symbol in the panel fallback text and `decision_model`'s reason string; fix the doubled word.
- **Depends on:** 10 (the log schema is declared against the contract first). **Unblocks:** a backtest-gate item (Item 6); the release gate if Item 6 is adjudicated Critical.
- **Effort:** medium.
- **Risk of doing:** log schema churn if fields change later — mitigated by the T2-3 declaration; disk growth is trivial.
- **Risk of not:** every panel asserts a log that does not exist; "why this decision" stays unanswerable after the process exits; Item 6 blocks the backtest gate indefinitely.

13. Item 14 — Risk Is Not Conviction *(+ Items 10(b)/(c) merged; Entry #25 riders)*

- **Position:** After 11 because the shared validation-gate seam lands there; after 12 per the stricter-reading order (swap if Item 6 is adjudicated Major). Remove the `risk_score`/`signal_strength` aliases (one defect reported under both Item 10(c) and Item 14(a)); remove `bias_factor` from `stop_mult` — conviction currently tightens stops arithmetically, which is Item 14's prohibition written as code and reaching the STOP LOSS line; wire `volatility_state` into the `calculate_stop_targets` call — engine_core omits the parameter, so the HIGH/EXTREME widening tiers are permanently inert (verified in source; recorded in Entry #25 from the truncated first audit attempt); normalize entry-zone lower/upper at source instead of exporting inverted bounds whenever EMA20 > EMA50. Riders: the `eq_trade_direction` LONG-by-default scoring of no-signal runs; `live_trading`'s panel-dump side effect (an interface question — belongs with contracts).
- **Depends on:** 10, 11. **Unblocks:** honest stop geometry on the panel.
- **Effort:** small-medium.
- **Risk of doing:** stop distances change (widen under high volatility, decouple from conviction) and risk_valid/NO-TRADE distributions shift — intended, but a flipped entry decision is a real behavior change to be accepted knowingly via scan diff, not silently.
- **Risk of not:** conviction keeps laundering into risk geometry and the decision gate — the ergodicity rationale's exact concern, live on the panel every run.

14. T2-4 — Explicit configuration *(+ cosmetic riders)*

- **Position:** After 5 (do not wire knobs to indicators just deleted) and after 11/13 (the decision thresholds being named just changed meaning). Wire surviving indicator lengths and volume-profile bins through config; move the decision thresholds (75/70, ±30/±20) to named constants with one reading point; chart dimensions/dpi/style through config — note config says 150 dpi/height 10 while plotting hardcodes 200/8, so wiring changes chart artifacts (intended, visible). Rider: candle body width (0.0008 days is invisible on a 4h chart — cosmetic, unrated; fix while touching plotting).
- **Depends on:** 5, 11. **Unblocks:** config actually configures; T2-4's own verification (change a knob, rerun pinned, observe the intended delta) becomes executable — it was impossible before item 3 existed.
- **Effort:** small.

- **Risk of doing:** near-zero on the decision path (config values equal current hardcodes) — assert via golden invariance; chart files change, which is visible but intended.
- **Risk of not:** the file keeps advertising central control the code ignores — every future parameter change is a trap.

15. Item 2's standing caveat — bfill quarantine + point-in-time harness

- **Position:** Last code item because nothing in the live path reads it (the decision row is always the last bar; backward-fill cannot reach it), because its verification apparatus — re-computing historical decisions on truncated data — exists in no form and is itself the on-ramp to Step 9, and because the backtest gate is the first consumer. Remove or quarantine backward-fill in analysis paths; build the truncated-dataframe check the finding names. If Viktor upholds Claude's dispute (Compliant → Non-compliant), this becomes formally mandatory for the backtest gate — same work, same position; the dispute changes force, not order. Note: the backtest gate's letter (Items 2, 3, 6, 18) does not require Items 11/13 fixed — this plan fixes them earlier anyway, because backtesting an engine with double-counted confidence manufactures precisely the false validation Item 7 exists to prevent.
- **Depends on:** 4, 7, 8. **Unblocks:** backtesting, honestly.
- **Effort:** medium.
- **Risk of doing:** removing bfill surfaces NaNs the fills were hiding → halts — item 9's fail-safe path doing its job; detect via suite.
- **Risk of not:** the backtest gate opens onto a look-ahead leak — the Constitution's own words for the most common way a system convinces itself it works.

16. Step 8 re-audit — independent, scoped to changed items

- **Position:** Terminal, per the ratified sequence: re-audit belongs to the independent auditor, not the fixer. Scope it to changed items — cheap under the pay-per-use cost structure in Entry #13 — with artifact evidence (panel screenshots, test output, log files), per the schema's Evidence discipline. This is where the release gate formally opens (three confirmed Criticals resolved-and-verified; Item 6 as well if adjudicated Critical) and where the backtest-gate condition is checked. Close what the remediation generated evidence for: T3-1/T3-2/T3-8 (commit-per-change records, dated notes, recorded hashes), T3-5 (item 3). Item 15 stays Unknown until backtesting exists — the honest status.
- **Depends on:** all. **Unblocks:** reliance on output; Step 9.
- **Effort:** small (process).
- **Risk of doing:** none code-wise; the process risk is treating it as a formality — the Constitution's own warning that agreement is not validation.
- **Risk of not:** self-certification — the exact conflict the audit architecture exists to prevent.

Placement of the three harness results

- **Clone failure → severity correction on T2-6 (item 2).** It is the same defect the blind run found and Run C graded; the correction is that the manifest gap does not degrade gracefully in practice — it prevents startup outright. It belongs to T2-6 rather than Item 13 (Fail Safely) because the failure occurs at install/import, before any analysis exists that could fail safely.
- **Hardcoded asset name → supplementary finding under Item 6, carried by item 12.** The explanation layer must trace to the run it explains; naming the wrong asset severs output→run lineage, and it misleads on every non-AERO run, which the scan report shows is normal use. It was never rated because no register run examined explanation-text provenance.
- **Doubled word → Minor cosmetic, no Tier 1 anchor (closest: T4-3), carried by item 12.** It rides that item because the same strings are edited there. Both unrated findings were registered at step 0 so they cannot silently vanish — the harness earned that discipline by catching what four graded passes missed.

Do the disputes change the sequencing?

- **Item 6's severity — yes, conditionally.** It decides whether item 12 precedes item 13 (stricter reading → sequenced as shown) and it changes the minimal gate-opening set (Q5). Everything else is invariant to the ruling — which is deliberate: the plan is built to be optimal under the stricter reading and merely valid under the weaker one.
- **Item 2's strength — no reorder.** The caveat's own required action already pins this work immediately before backtesting; the dispute changes its force (advisory → mandatory) and adds a re-audit obligation if upheld.
- **Items 4/12 — no reorder, dissolved by repair.** Deleting the cache at item 6 makes both readings true; repair is cheaper than argument and spends none of Viktor's adjudication budget. If a Non-compliant ruling lands anyway, the fix has already shipped.
- Consequently: request the rulings at step 0, because two of them touch ordering.

The five questions

1. Which fixes cannot be verified once made, given what currently exists?

Items 3, 11, and 13 — their Verification fields demand corrupted-data fixtures, fault injection, and ablation runs, and none of that apparatus exists (no test framework, no pinned dataset, no path to feed one into the engine). Also the Item 4/12 cache hazard (needs a same-process double-fetch test), T2-4's knob-effect check (impossible while every run fetches different live data), T2-3's mypy gate (mypy absent), and Item 2's caveat (a point-in-time harness — nothing close exists; it is half a backtester). **Implication:** all of these sit after the apparatus phase, which is therefore not process hygiene but the gate-opener — "resolved" per the finding schema requires Verification, and the release gate requires resolution. A fix made before its apparatus exists can be *made*, never *resolved*. The deeper point: verification-by-reading is precisely the shared-habits trap the Constitution's authorship-versus-judgment-independence distinction names; artifacts, not arguments. And one residual even after the apparatus exists: Item 11's tests prove the change matched intention, not that the intention is right — only the independent re-audit closes that.

2. Which fixes change what the trader sees?

Visible: item 8 (new rejection/error states; UNAVAILABLE macro; possibly a different current price under the forming-candle policy), item 9 (halts and error panels instead of confident fabrications), item 11 (confidence and validation numbers drop; MOMENTUM and Current-Market lines disappear), item 13 (stop distances move; entry zone reads correctly; VALIDATION's meaning changes), item 12 (true log line, real chart path, version stamp, Item 7 label, symbol-correct text), item 14 (chart dimensions/dpi). Invisible: items 1, 3, 4, 6, 7, 10, and 15 on the live path (item 2 is visible only as the absence of a crash). **Does it matter for ordering?** Yes, three ways — but not as the primary key. (a) The shut release gate makes *now* the cheapest possible moment for visible churn: all visible changes are sequenced inside the window, and the gate opens only after the panel has settled — post-gate number drift is the trust-erosion scenario. (b) Visible changes are self-evidencing: the finding schema lists panel screenshots as valid Evidence, so visible fixes carry the cheapest audit trail. (c) The coupling rule at item 11: a number and the prose explaining it must change in one commit, or the panel lies in words while telling the truth in numbers. Dependencies and verifiability set the top-level order; visibility is a batching and timing constraint.

3. Are any two findings the same underlying defect reported twice?

Four merges. (1) The phantom trade-log claim: one defect, reported under Items 5, 6, and 8(b) — one work item (12) satisfying all three. (2) The fabricated fallback constants: one defect family under Item 3's fabrication clause, Item 8(a), Item 13, and blind-run finding 2 — one work item (9), while Item 3's validator remains separate scope (8). (3) The `risk_score`/`signal_strength` aliasing: Item 10(c) and Item 14(a) are the same edit, made once (13). (4) trend_health's overloading: Item 11's scoring double-count and Item 10(a)'s triple rendering are one root defect — one quantity doing four jobs — fixed in one controlled change (11); and Item 14(b)'s validation-gate rebasing is the same edit as Item 11's, sequenced once. Consequence: Item 8 has no unique remediation work — it is the parent principle whose every concrete instance is co-reported; its re-audit rides items 9 and 12. The invariants remain distinct for re-audit even where the edits merge. *Correlation flag:* merging findings is interpretation, and I share habits with the reporters; the re-audit should confirm the merges, and a wrong merge is cheap to split because the underlying edits are separately committable.

4. Is any Non-compliance cheaper to fix by deleting than by repairing?

Five. (a) Item 16's dead indicators — deletion is strictly cheaper; wiring them would be new-feature work. (b) `compute_exit` — delete rather than wire: wiring would change decision semantics and presupposes a held-position concept the engine deliberately lacks, while exit_watch already performs the advisory role. (c) The indicator/structure caches — delete rather than repair the key: they never hit across runs, and deletion removes both the stale-serving hazard and the mutation hazard at zero functional cost. (d) The fabricated fallbacks — deleting them in favour of the existing error path is cheaper than building a degraded-mode subsystem, and satisfies Item 13's "halt output" branch; the flag-and-degrade route is the auditor's letter and the richer repair, and the choice between them is Viktor's precisely because my simplicity preference is correlated with the builder's (flagged at item 9). (e) The MOMENTUM/Current-Market renderings — delete the redundant lines rather than invent distinct underlying metrics. All five deletions are git-reversible and proven safe by invariance checks; none touches anything with a consumer (VWMA stays — `entry_model` consumes it).

5. What is the smallest set of changes that would let the release gate open?

The T2-6 clone fix (outside the gate's letter, inside its spirit — nothing is verifiable from a fresh clone without it); the thin apparatus slice — pinned dataset, corrupted-data fixtures, fault injection, one golden baseline — which is the executable minimum of the three Criticals' Verification fields; the Item 3 fix; the Item 13 fix (which carries most of Item 8's shared defect with it); the Item 11 fix; the Item 6 fix under the conservative reading — or Viktor's Major ruling, which removes it at zero code cost, making the adjudication itself part of the minimal path; then the scoped independent re-audit of exactly those items, since "resolved" is not self-declared. Not in the minimal set: Items 5, 7, 10, 14, 16, T2-1/3/4, the full T3-4 build, T3-7 — with two honesty caveats: the phantom "Trade logged" line is a one-line deletion whose exclusion would leave the panel lying on the very first relied-upon run, so include it regardless; and opening the release gate is not the backtest gate — the latter additionally requires Item 6 fully Compliant, Item 2's caveat honored (item 15), and the Tier-3 safety net actually real (items 1, 3, 4, 7), and this plan deliberately fixes more than that gate's letter demands, because a backtest run against an engine with inflated confidence and unwired volatility tiers would validate a fiction.